

In C Program

* Weak 1 Module

1) Syntax:-

- It is the set of rules that defines the structure and format of a programming language
- If syntax rules are violated, the compiler gives an error

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    printf("Hello");
```

```
    return 0;
```

```
}
```

Rules of Syntax

- 1) ; (semicolon) ends every statement
- 2) { } (Curly braces) defines a block of code.
- 3) C is a Case sensitive
- 4) Execution starts from main()
- 5) Comments :-

// Single line

```
/*
```

```
    This is a
```

```
    multi-line comment
```

```
*/
```

2) What are condition and loops

8) Condition and loops are used to control the flow of execution

a) condition

```
• if (condition)
{
    // executes if condition is true
}
```

```
• else if (condition)
{
    // checks next condition
}
```

```
• else
{
    // executes if all condition are false
}
```

ex:-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int age = 18;
```

```
    if (age >= 18)
```

```
    {
```

```
        printf("Adult");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Minor");
```

```
    }
```

```
    return 0;
```

```
}
```


Loops

Loops are used to repeat a block of code multiple times

```
for (initialization; condition; update)
{
    // code
}
```

ex:-

```
#include <stdio.h>
```

```
int main()
{
```

```
    int i;
```

```
    for (i=1; i<=5; i++)
```

```
    {
```

```
        printf("%d\n", i);
```

```
    }
```

```
    return 0;
```

```
}
```

Note (Nested loops)

ex:-

```
for (i=1; i<=rows; i++) // outer loops -> Rows
```

```
{
```

```
    for (j=1; j<=cols; j++) // inner loops -> columns
```

```
    {
```

```
        // code
```

```
    }
```

```
}
```

~~* In C~~

1) Square

~~Example~~

```
#include <stdio.h>
```

```
int main() {
```

```
    int size;
```

```
    printf("Enter the size of the square:");
```

```
    scanf("%d", &size);
```

```
    for (int i = 0; i < size; i++) {
```

```
        for (int j = 0; j < size; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
* * *
* * *
* * *
* * *
* * *
```

2) Hollow square

```
#include <stdio.h>
```

```
int main() {
```

```
    int size;
```

```
    printf("Enter the size of the square:");
```

```
    scanf("%d", &size);
```

```
    for (int i = 1; i <= size; i++) {
```

```
        for (int j = 1; j <= size; j++) {
```

```
            if (i == 1 || i == size || j == 1 || j == size) {
```

```
                printf("*");
```

```
            } else {
```

```
                printf(" ");
```

```
            }
```

```
        printf("\n");
```

```
* * * * *
*           *
*           *
*           *
* * * * *
```


return 0;

3) Triangle

```
#include <stdio.h>
```

```
int main () {
```

```
    int rows = 5
```

```
    for (int i = 0; i < rows; i++) {
```

```
        for (int j = 0; j <= i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
*
* *
* * *
* * * *
* * * * *
```

4) Inverted Triangle

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int rows = 5;
```

```
    for (int i = 0rows; i <= rows1; i++)
```

```
    {
```

```
        for (int j = 0; j <= rowsrows - i; j++)
```

```
        {
```

```
            printf(" ");
```

```
* * * * *
* * * *
* * *
* *
*
```

```

    }
    printf("\n");

```

```

}
return 0;

```

5) Pyramid

```

      *
     * *
    * * *
   * * * *
  * * * * *

```

```

#include <stdio.h>

```

```

int main()

```

```

{

```

```

    int rows = 5;

```

```

    for (int i = 0; i <= rows; i++)

```

```

    {

```

```

        for (int j = 0; j <= (rows - i); j++)

```

```

        {
            printf(" ");

```

```

        }

```

```

        for (int k = 0; k <= 2 * i - 1; k++)

```

```

        {

```

```

            printf("* ");

```

```

        }

```

```

        printf("\n");

```

```

    }

```

```

    return 0;

```

```

}

```


inverted full pyramid

```
* * * * *
  * * * *
    * * *
      * *
        *
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int rows = 5;
```

```
    for (int rows = rows; rows > 0; rows--)
```

```
    {
```

```
        for (int j = 0; j < rows; j++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        for (int j = 0; j < 2 * rows - 1; j++)
```

```
        {
```

```
            printf("* ");
```

```
        }
```

```
        printf("\n");
```

```
    }  
    return 0;
```

7) Half diamond

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int n=5;
```

```
    for(int i=1; i<=n; i++) {
```

```
        for(int j=1; j<=i; j++)
```

```
            cout << " * ";
```

```
        cout << "\n";
```

```
    }
```

```
    for(int i=n-1; i>=1; i--) {
```

```
        for(int j=1; j<=i; j++)
```

```
            cout << " * ";
```

```
        cout << "\n";
```

```
    }
```

```
    return 0;
```

```
}
```

```

 *
 * *
 * * *
 * * * *
 * * * * *
 * * * *
 * * *
 * *
 *

```



```

      *
    * * *
  * * * * *
* * * * * *
  * * * *
    * * *
      * *
        *
  
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n=5;
```

```
    for (int i=0; i < 2 * n - 1; i++)
```

```
{
```

```
    int comp;
```

```
    if (i < n)
```

```
        comp = 2 * (n - i) - 1;
```

```
    else
```

```
        comp = 2 * (i - n + 1) + 1;
```

```
    for (int j=0; j < comp; j++)
```

```
        printf(" ");
```

```
    for (int k=0; k < 2 * n - comp; k++)
```

```
        printf(" * ");
```

```
    printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

Main.c x +

```
1  #include <stdio.h>
2
3  int main() {
4      int size;
5
6      printf("Enter the size of the square: ");
7      scanf("%d", &size);
8
9      for (int i = 0; i < size; i++) {
10         for (int j = 0; j < size; j++) {
11             printf("*");
12         }
13         printf("\n");
14     }
15
16     return 0;
17 }
```

Console

I/O

AI Agent

5.2s

Enter the size of the square: 5

```
*****
*****
*****
*****
*****
```


Main.c × +

```
1 #include<stdio.h>
2 int main(){
3     int size;
4     printf("enter the size of the square:");
5     scanf("%d",&size);
6     for(int i=1; i<=size; i++){
7         for(int j=1; j<=size; j++){
8             if(i==1||i==size||j==1||j==size){
9                 printf("*");
10            }else{
11                printf(" ");
12            }
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

Console

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enter the size of the square:5

* *

* *

* *

C Main.c x +

```
1 #include <stdio.h>
2
3 int main() {
4     int rows = 5;
5
6     for (int i = 1; i <= rows; i++) {
7         for (int j = 1; j <= i; j++) {
8             printf("* ");
9         }
10        printf("\n");
11    }
12
13    return 0;
14 }
```

Console

I/O

AI Agent

```
*
* *
* * *
* * * *
* * * * *
```



C Main.c x +



```
1  #include <stdio.h>
2
3  int main() {
4      int rows = 5;
5
6      for (int i = rows; i >= 1; i--) {
7          for (int j = 1; j <= i; j++) {
8              printf("* ");
9          }
10         printf("\n");
11     }
12
13     return 0;
14 }
```

Console

I/O

AI Agent

218 ms

```
* * * * *
* * * *
* * *
* *
*
```




C Main.c x +



```
1  #include <stdio.h>
2
3  int main() {
4      int rows = 5;
5
6      for (int i = 1; i <= rows; i++) {
7
8          // Print spaces
9          for (int j = 1; j <= rows - i; j++) {
10             printf(" ");
11         }
12
13         // Print stars
14         for (int j = 1; j <= 2 * i - 1; j++) {
15             printf("*");
16         }
17
18         printf("\n");
19     }
20
21     return 0;
22 }
```

Console

I/O

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231 ms ✓

```
*
***
*****
*****
*****
```



Main.c x +



```
1  #include <stdio.h>
2
3  int main() {
4      int rows = 5;
5
6      for (int i = rows; i >= 1; i--) {
7
8          // Print spaces
9          for (int j = 1; j <= rows - i; j++) {
10             printf(" ");
11         }
12
13         // Print stars
14         for (int j = 1; j <= 2 * i - 1; j++) {
15             printf("*");
16         }
17
18         printf("\n");
19     }
20
21     return 0;
22 }
```

Console

I/O

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195 n

```
*****
*****
*****
***
*
```

Main.c

+

Console

I/O

AI Agent

228 ms

```
1 #include <stdio.h>
2
3 int main() {
4     int rows = 5;
5
6     // Upper half
7     for (int i = 1; i <= rows; i++) {
8         for (int j = 1; j <= i; j++) {
9             printf("* ");
10        }
11        printf("\n");
12    }
13
14    // Lower half
15    for (int i = rows - 1; i >= 1; i--) {
16        for (int j = 1; j <= i; j++) {
17            printf("* ");
18        }
19        printf("\n");
20    }
21
22    return 0;
23 }
24
```

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```




C Main.c × +



```
1  #include <stdio.h>
2
3  int main() {
4      int rows = 5;
5      // Upper pyramid
6      for (int i = 1; i <= rows; i++) {
7          // Print spaces
8          for (int j = 1; j <= rows - i; j++) {
9              printf(" ");
10         }
11         // Print stars
12         for (int j = 1; j <= 2 * i - 1; j++) {
13             printf("*");
14         }
15         printf("\n");
16     }
17     // Lower inverted pyramid
18     for (int i = rows - 1; i >= 1; i--) {
19         // Print spaces
20         for (int j = 1; j <= rows - i; j++) {
21             printf(" ");
22         }
23         // Print stars
24         for (int j = 1; j <= 2 * i - 1; j++) {
25             printf("*");
26         }
27         printf("\n");
28     }
29     return 0;
30 }
```

Console

I/O

AI Agent

```
      *
      ***
      *****
      *******
      *********
      *****
      *****
      ***
      *
```



In C++ Program

1) Syntax:- Syntax is ~~the~~ the set of rules that defines the structure and format of a programming language.

- If syntax rules are violated compiler gives error

```
#include <iostream>
```

```
using namespace std;
```

```
int main () {
```

```
    cout << "Hello";
```

```
    return 0;
```

```
}
```

Rules of Syntax

- a) semi colon ends every statements
- b) { } curly brackets define block of code
- c) C++ is case sensitive
- d) Execution starts from main() only
- e) for comments: // single line
 L multi line - 1

2) what are condition and loops

Condition and loops are used to control flow of execution

a) condition

- if (condition) { }

- ↳ if true ; execute block

- else if (condition) { }

- ↳ check next condition

?

se 3

↳ if all false; execute this

1) Loops

Read code n times for loop syntax
for (initialization; condition; update)

Note;

Outer loop → Row

Inner loop → (column)

Programs

1) square

```
#include <iostream>
using namespace std;
```

```
int main () {
```

```
    int n = 5;
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < n; j++) {
```

```
            cout << "X ";
```

```
        cout << endl;
```

```
    return 0;
```

```
}
```

```
X X X X X
X X X X X
X X X X X
X X X X X
X X X X X
```

3

3

2) Hollow square

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
void hollowSquare (int rows)
```

```
{
```

```
    int i, j;
```

```
    for (i=1; i<=rows; i++)
```

```
    {
```

```
        if (i==1 || i==rows)
```

```
            for (j=1; j<=rows; j++)
```

```
                cout << " * ";
```

```
        else
```

```
            for (j=1; j<=rows; j++)
```

```
                if (j==1 || j==rows)
```

```
                    cout << " * ";
```

```
            else
```

```
                cout << "  ";
```

```
        cout << "\n";
```

```
    }
```

```
}
```

```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```


Example

```
#include <iostream>
```

```
using namespace std;
```

```
void pypart (int n)
```

```
{
```

```
    for (int i=0; i<n; i++) {
```

```
        for (int j=0; j<=i; j++) {
```

```
            cout << " * ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
}
```

*

* *

* * *

* * * *

* * * * *

4) Inverted Triangle

```
#include <iostream>
using namespace std;
void pypart(int n)
```

```
{
```

```
    for (int i = n; i > 0; i--) {
```

```
        for (int j = 0; j < i; j++) {
```

```
            cout << " * ";
```

```
        }
```

```
        cout << endl;
```

```
}
```

* * * * *

* * * *

* * *

* *

*

pyramid

```
#include <iostream>
using namespace std;
```

```
void py part 2 (int n)
{
```

```
    int i, j, k = n;
```

```
    for (i = 1; i <= n; i++) {
```

```
        for (j = 1; j <= n; j++) {
```

```
            if (j >= k)
```

```
                cout << " * ";
```

```
            else
```

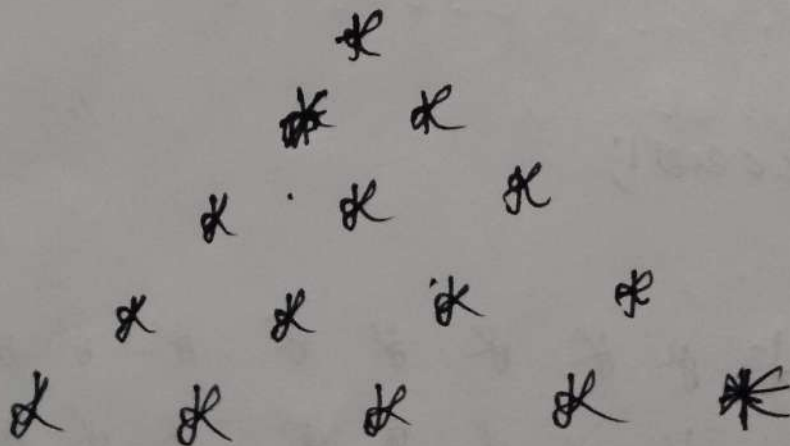
```
                cout << "  ";
```

```
        }
```

```
        k--;
```

```
        cout << "\n";
```

```
    }
```



b) Inverted Pyramid

```
#include <iostream>
```

```
using namespace std;
```

```
void print InvTriangle (int n)
```

```
{
```

```
    for (int i=0; i<n; i++) {
```

```
        int space = i;
```

```
        for (int j=0; j<2*n-i-1; j++) {
```

```
            if (space) {
```

```
                cout << "  ";
```

```
                space --;
```

```
            }
```

```
            else {
```

```
                cout << "x ";
```

```
            }
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
}
```

```
x x x x x x x x x x
```

```
x x x x x x x x
```

```
x x x x x x
```

```
x x x x x
```

```
x x x
```

```
x
```


half diamond

```
#include <iostream>
using namespace std;
void printRotatedTriangle (int n)
{
```

```
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
```

```
            cout << "*" << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    for (int i = 1; i <= n-1; i++) {
```

```
        for (int j = 1; j <= n-i; j++) {
```

```
            cout << " " << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
}
```

```
*
**
***
****
*****
****
***
**
*

```

8) full diamond

#include <iostream>

using namespace std;

void Diamond (int n)

{

for (int i=1; i<=n; i++) {

for (int j=1; j<=n-i; j++) {

cout << " ";

for (int k=1; k<=2*i-1; k++) {

cout << " * ";

}

cout << endl;

}

// Now we print other (n-1) rows of the pattern

for (int i=1; i<=n-1; i++) {

for (int j=1; j<=i; j++) {

cout << " ";

}

for (int k=1; k<=2*(n-i)-1; k++) {

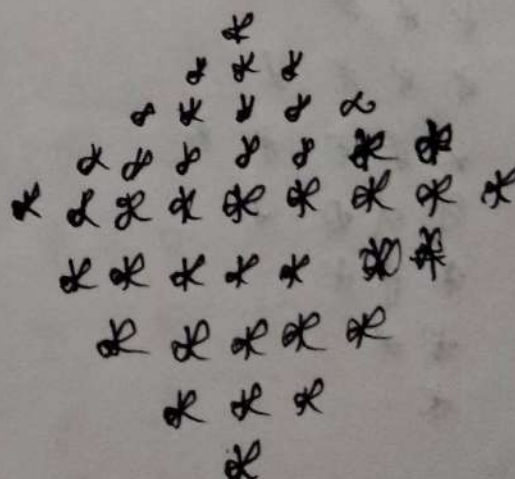
cout << " * ";

}

cout << endl;

}

}



Main.cpp x +

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6
7      for (int i = 1; i <= n; i++) {
8          for (int j = 1; j <= n; j++) {
9              cout << "* ";
10             }
11             cout << endl;
12         }
13
14         return 0;
15     }
```

Console

I/O

AI Agent

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```




Main.cpp × +



```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter the size of the square: ";
7      cin >> n;
8
9      for (int i = 1; i <= n; i++) {
10         for (int j = 1; j <= n; j++) {
11
12             // Print '*' on the border
13             if (i == 1 || i == n || j == 1 || j == n)
14                 cout << "* ";
15             else
16                 cout << " "; // Print spaces inside
17         }
18         cout << endl;
19     }
20
21     return 0;
22 }
```

Console

I/O

AI Agent

Enter the size of the square: 5

```
* * * * *
*       *
*       *
*       *
* * * * *
```

<>

OneCompiler

Upgrade

AI

C++

Run

Main.cpp

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number of rows: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "* ";
        }
        cout << endl;
    }

    return 0;
}
```

Console

I/O

AI Agent

Enter number of rows: 5

*

* *

* * *

* * * *

* * * * *

Main.cpp × +

Console

I/O

 AI Agent

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number of rows: ";
7      cin >> n;
8
9      for (int i = n; i >= 1; i--) {
10         for (int j = 1; j <= i; j++) {
11             cout << "* ";
12         }
13         cout << endl;
14     }
15
16     return 0;
17 }
```

```
Enter number of rows: 5
* * * * *
* * * *
* * *
* *
*
```


Main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number of rows: ";
7      cin >> n;
8
9      for (int i = 1; i <= n; i++) {
10
11         // Print spaces
12         for (int j = 1; j <= n - i; j++) {
13             cout << " ";
14         }
15
16         // Print stars
17         for (int j = 1; j <= 2 * i - 1; j++) {
18             cout << "*";
19         }
20
21         cout << endl;
22     }
23
24     return 0;
25 }
```

Console

I/O

AI Agent

Enter number of rows: 5

```
*
 ***
*****
*****
*****
```

Main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number of rows: ";
7      cin >> n;
8
9      for (int i = n; i >= 1; i--) {
10
11         // Print leading spaces
12         for (int j = 1; j <= n - i; j++) {
13             cout << " ";
14         }
15
16         // Print stars
17         for (int j = 1; j <= 2 * i - 1; j++) {
18             cout << "*";
19         }
20
21         cout << endl;
22     }
23
24     return 0;
25 }
```

Console

I/O

AI Agent

Enter number of rows: 5

*

Main.cpp × +

Console

I/O

AI Agent

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number of rows: ";
7      cin >> n;
8
9      // Upper half
10     for (int i = 1; i <= n; i++) {
11         for (int j = 1; j <= i; j++) {
12             cout << "* ";
13         }
14         cout << endl;
15     }
16
17     // Lower half
18     for (int i = n - 1; i >= 1; i--) {
19         for (int j = 1; j <= i; j++) {
20             cout << "* ";
21         }
22         cout << endl;
23     }
24
25     return 0;
26 }
```

Enter number of rows: 5

```
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* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Main.cpp

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cout << "Enter number of rows: ";
6      cin >> n;
7      // Upper pyramid
8      for (int i = 1; i <= n; i++) {
9          // Print spaces
10         for (int j = 1; j <= n - i; j++) {
11             cout << " ";
12         }
13         // Print stars
14         for (int j = 1; j <= 2 * i - 1; j++) {
15             cout << "*";
16         }
17         cout << endl;
18     }
19     // Lower inverted pyramid
20     for (int i = n - 1; i >= 1; i--) {
21         // Print spaces
22         for (int j = 1; j <= n - i; j++) {
23             cout << " ";
24         }
25         // Print stars
26         for (int j = 1; j <= 2 * i - 1; j++) {
27             cout << "*";
28         }
29         cout << endl;
30     }
31     return 0;
32 }
```

Console

I/O

AI Agent

Enter number of rows: 5

```
*
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*****
```